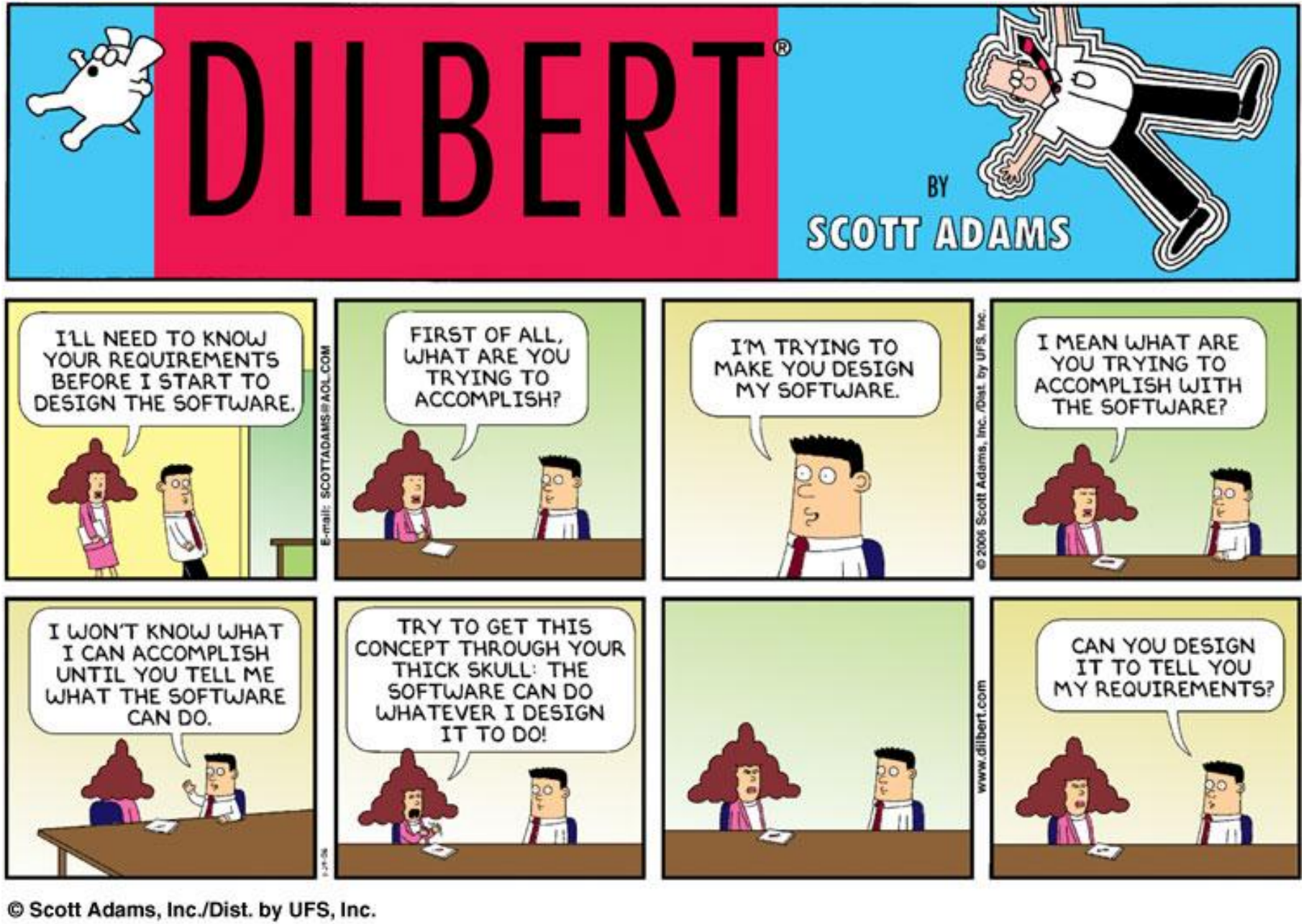
Requirements Elicitation and User Stories

Lecture 4

Agenda

- Techniques for Requirements Elicitation and Analysis
  - Ethnography
  - Interviews
  - Questionaries
  - Workshops of Requirements
  - Prototypes
- Describing and Documenting Requirements: User Stories
  - Story Mapping
  - Criteria for Good Stories

Techniques for Requirements Elicitation and Analysis



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Ingeniería de requerimientos

- Consecuencias de una pobre definición de requerimientos:
  - La mayor causa de proyectos fallidos
  - Principales causas de fallas de sistemas:
    - ❖Requerimientos y especificaciones incompletas
    - ❖Requerimientos y especificaciones cambiantes
    - ❖Falta de aporte del usuario
    - ❖Reporte caótico de la información
  - Calidad, costos y tiempos de entrega del producto de software se ven afectados

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Ventajas de requerimientos bien especificados

- Sistemas de calidad satisfacen las necesidades de los stakeholders
- Los errores en requerimientos son más costosos de corregir a medida que tardan en ser detectados
- Los cambios de la mayoría de requerimientos durante el desarrollo incrementan el costo del producto final

Proceso de la Ingeniería de Requerimientos

- Elicitación de requerimientos: técnica para descubrir los requerimientos del usuario y stakeholders
- Análisis de requerimientos: Registrar los requerimientos en una forma clara, concisa, y no ambigua.
  - Texto
  - Modelos de sistemas
- Chequeo: identificar y resolver contradicciones en los requerimientos.
- Validación de requerimientos: revisar la calidad de los requerimientos.
- Manejo de requerimientos
  - Organizar la documentación de requerimientos para comunicar efectivamente los requerimientos a los stakeholders
  - Organizar los requerimientos para lidiar con el cambio.

Proceso de la Ingeniería de Requerimientos

- Elicitación
  - Descubrir los requerimientos del dominio de la aplicación
- Análisis y especificación
  - Definir los servicios provistos
  - Definir las restricciones
  - Clasificarlos
  - Identificar criterios de validación
- Chequeo
  - Identificar y resolver conflictos, ambiguedades, requerimientos innecesarios
  - Un proceso complejo de negociación entre los stakeholders
- Validación
  - Revisar y llegar a acuerdos
- Manejo de requerimientos
  - Registrar todo en los documentos de requerimientos
  - Manejar el cambio a los requerimientos

Técnicas de elicitación

1. Etnografía (también para entender el dominio)
2. Entrevistas
3. Cuestionarios
4. Taller de Requerimientos
5. Prototipos

Técnicas de Elicitación

- Estudiar el dominio del problema
  - El trabajo que se realiza en relación al alcance del proyecto
  - El trabajo que es realizado por medio de interacción con el sistema
- Determinar la forma en la cual el sistema debe responder a los eventos del negocio
- Identificar los requerimientos
- Definir las restricciones (limitaciones)/requerimientos no funcionales

Etnografía (1/5)

- Observar el proceso para entender el dominio.
- Puede ser:
  - Pasivo: Observar sin interrupción, video, recolectar y analizar documentación existente
  - Activo: participar en el trabajo mismo, tomando un rol.
- La mejor manera de entender la cultura y el dominio del problema

Etnografía

- Identifica requerimientos implícitos
- Descubre el proceso “real” en vez de el proceso documentado
- Identifica los problemas/eficiencias que no han sido previamente documentadas
- Identifica lo “obvio” que personas calificadas podrían no mencionar
- El ethnographer debe ser tratado como un “espíritu” y no tratar a las personas como “objetos de estudio”

Entrevistas (2/5)

- Una técnica muy común utilizada en casi todos los procesos de ingeniería de requerimientos para recopilación de hechos e información.

Entrevistas

- El conocimiento se transfiere desde los expertos del dominio
- Recopilación de requerimientos de los stakeholders
- Entrevistas estructuradas
- Entrevistas no estructuradas
- Evitar preguntas parcializadas, subjetivas, o impositivas.

Entrevistas

- Prepare la entrevista: No decir “Cuénteme lo que desee”
- Preséntese a usted mismo
- Pida permiso para tomar notas
- Escuche cuidadosamente, tenga tacto y ponga atención a los detalles.
- Sea paciente
- Investigue primero
- Use una lista de chequeo para la entrevista
- Pregunte si hay algo más que agregar
- Envíe un resumen de la entrevista al entrevistado

Lista de Chequeo de la Entrevista

- Establezca el perfil del stakeholder
  - Cargo, responsabilidades
- Evalúe el problema
  - ¿Por qué existe? ¿Cómo resolverlo?
- Entienda el entorno del stakeholder
  - ¿Quiénes son los stakeholders, niveles de experiencia, entrenamiento, expectativas de usabilidad, tecnología actual y futura
- Recapitule para verificar un buen entendimiento
- Utilice gráficos, diagramas para representar el problema del cliente
- Evalúe la solución
- Evalúe tipos de requerimientos no funcionales
- Resuma
  - Identifique los tres problemas o necesidades prioritarias

Cuestionarios (Encuestas) (3/5)

- Los cuestionarios no son un substituto para las entrevistas
  - Las preguntas no son adaptadas a la situación del stakeholder
  - No se puede identificar ni explicar asunciones
  - Complejidad para adaptar las preguntas a medida que la información es descubierta

Cuestionarios (Encuestas)

- Una buena técnica corroborativa de la mano con las entrevistas
- Rápida recopilación de datos provenientes de muchos stakeholders
- Use preguntas cerradas y preguntas de categorización/jerarquización (ranking)

Talleres de requerimientos (4/5)

- El análisis y la elicitación de requerimientos es un proceso social
- El grupo tiene un mejor entendimiento de los roles, responsabilidades, problemas y necesidades de todos.
- El enfoque de grupo provee una solución aceptable para todos los participantes.

Talleres de requerimientos

- El proceso del taller:
  - Se reúne un equipo de stakeholders y desarrolladores
  - Identificación del problema, consideración de soluciones, negociar, especificar requerimientos, llegar a acuerdos
  - Una reunión
  - Reglas para el taller y una agenda
  - Un facilitador
  - Una definición del mecanismo establecido

Prototipos (5/5)

- Los stakeholders no saben lo que quieren: la articulación del problema y la habilidad para visualizar el producto final
- Un prototipo de una versión inicial del sistema
- Se desarrolla rápidamente
- Reduce costos
- Efectivo para identificar requerimientos de interfaces de usuario

Prototipos

- Puede ayudar para validar el producto final
- Implementar el prototipo requiere un buen entendimiento de los requerimientos, a menudo identifican requerimientos incompletos o inconsistentes.
- Prototipos de papel o automatizados para simular funcionalidad

Describing and Documenting Requirements: User Stories in Scrum

What Are User Stories?

- User stories are a convenient format for expressing the desired business value for many types of product backlog items, especially features.
- User stories are crafted in a way that makes them understandable to both businesspeople and technical people.
- Ron Jeffries offers a simple yet effective way to think about user stories. He describes them as the three Cs: card, conversation, and confirmation.

Card

- The card idea is pretty simple. People originally wrote (and many still do) user stories directly on 3 × 5-inch index cards or sticky notes.

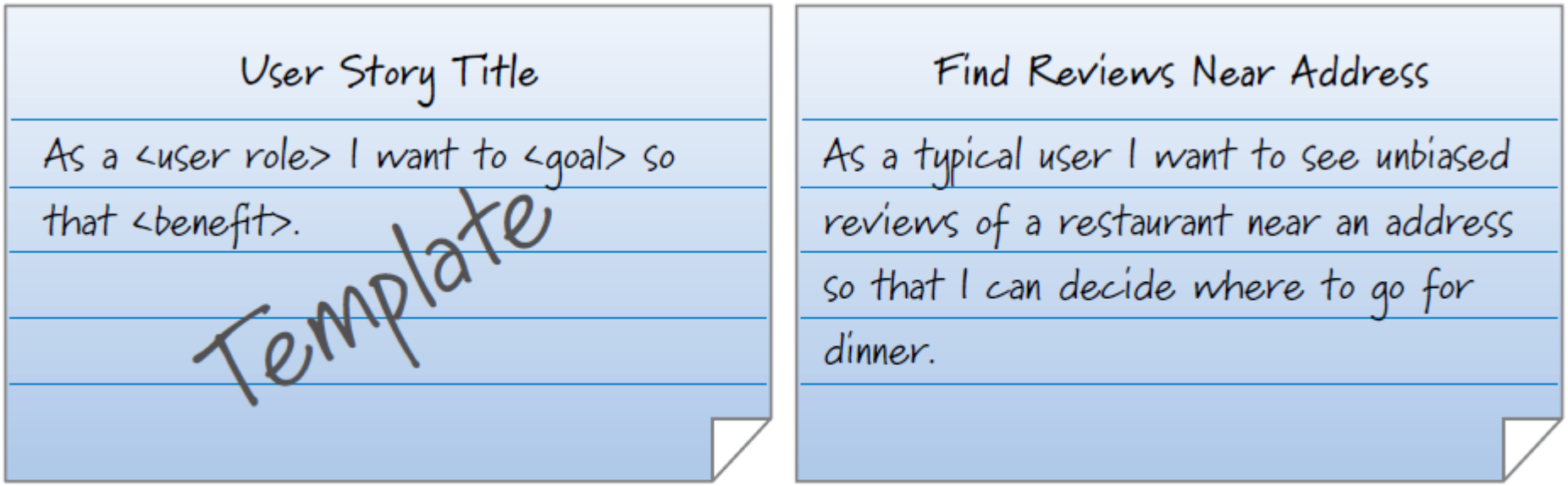


FIGURE 5.2 A user story template and card

Card

- The card isn’t intended to capture all of the information that makes up the requirement.
- In fact, we deliberately use small cards with limited space to promote brevity.
- A card should hold a few sentences that capture the essence or intent of a requirement.
- It serves as the placeholder for more detailed discussions that will take place among the stakeholders, product owner, and development team.

Conversation

- The details of a requirement are exposed and communicated in a conversation among the development team, product owner, and stakeholders.
- The user story is simply a promise to have that conversation.
- We say “that conversation,” but in actuality, the conversation is typically not a one-time event, but rather an ongoing dialogue.
- There can be
  - an initial conversation when the user story is written,
  - another conversation when it’s refined,
  - yet another when it’s estimated,
  - another during sprint planning (when the team is diving into the task-level details),
  - and finally, ongoing conversations while the user story is being designed, built, and tested during the sprint.

Conversation

- Although conversations are largely verbal, they can be and frequently are supplemented with documents.
- Conversations may lead to a UI sketch, or an elaboration of business rules that gets written down.

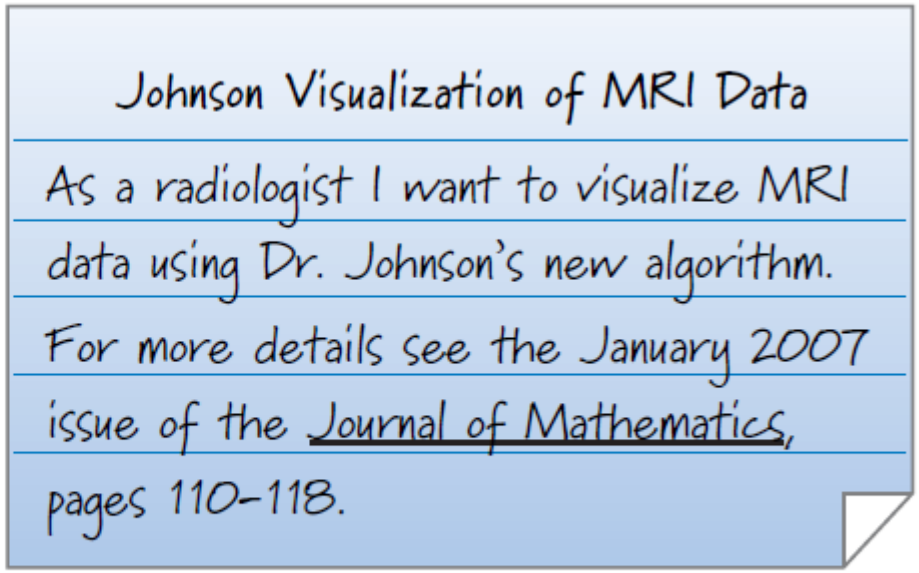


FIGURE 5.3 User story with additional data attached

Confirmation

- A user story also contains confirmation information in the form of conditions of satisfaction.
- These are acceptance criteria that clarify the desired behaviour.
- If the front of the card has a few-line description of the story, the back of the card could specify the conditions of satisfaction.
- These conditions of satisfaction can be expressed as high-level acceptance tests.

Confirmation

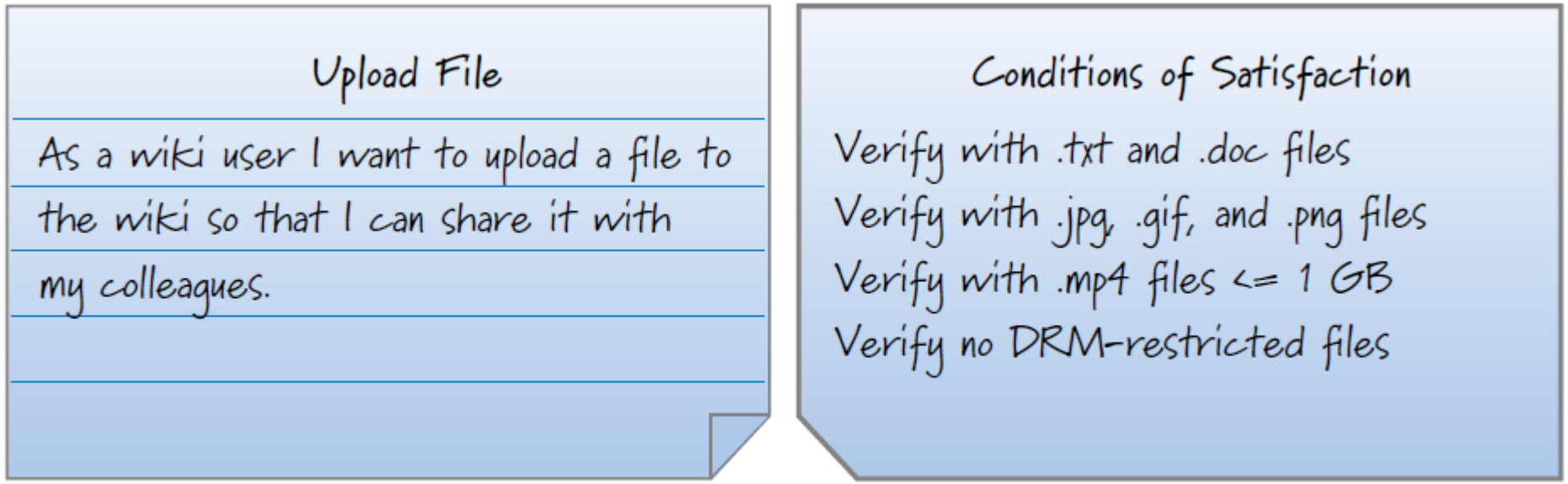


FIGURE 5.4 User story conditions of satisfaction

Level of Detail

- User stories can be written to capture customer and user needs at various levels of abstraction: epic, features, sprintable/implementable stories

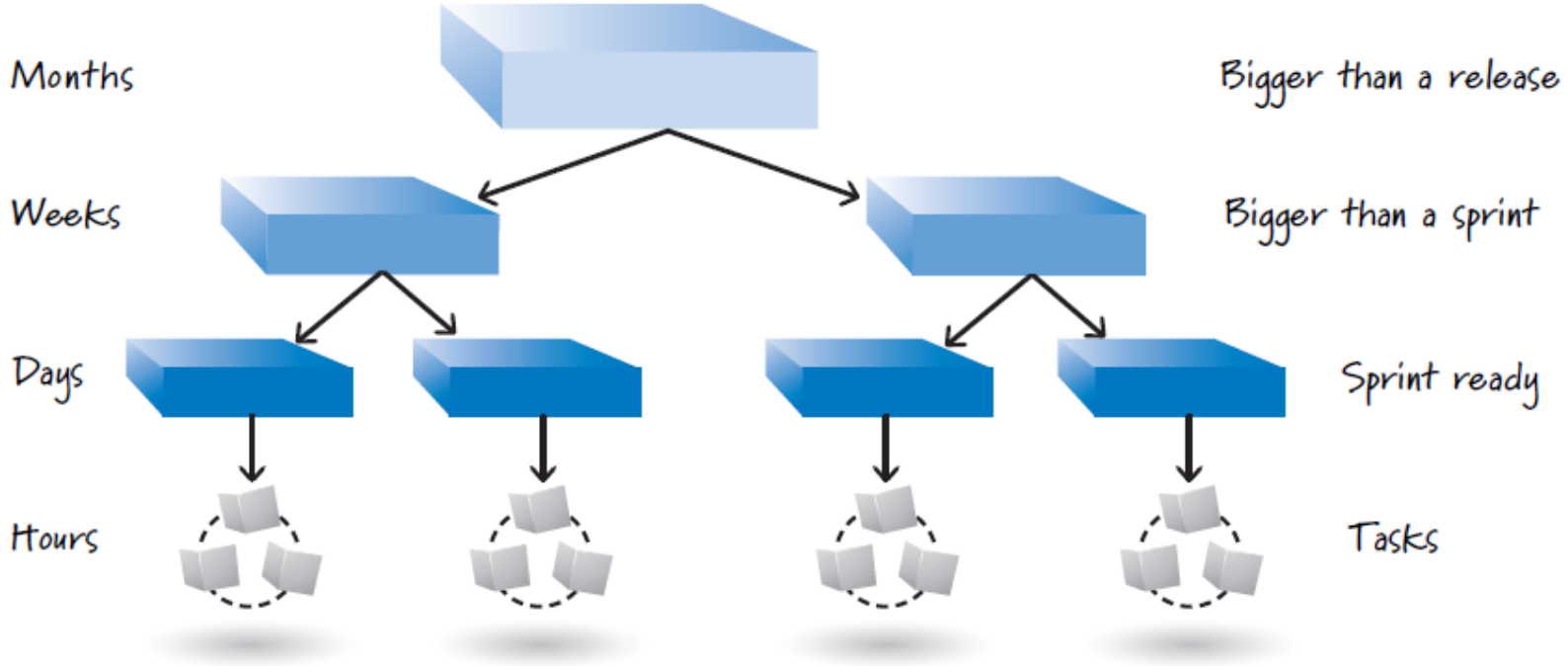


FIGURE 5.5 User story abstraction hierarchy

Level of Detail

- Some teams also use the term theme to refer to a collection of related stories.
- We may think of a theme as the summary card for a bunch of note cards stacked together with a rubber band around them to indicate that they are similar to one another in an area that we think is important.

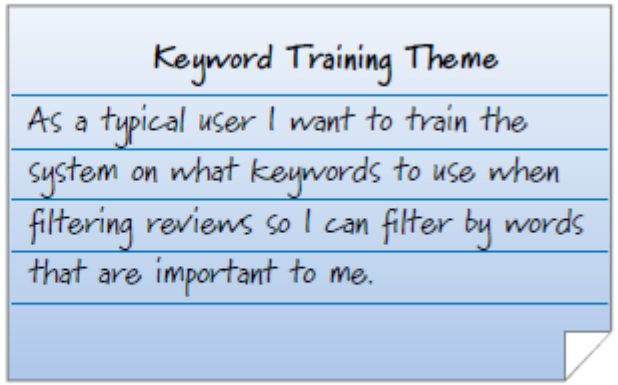


FIGURE 5.7 Example theme

INVEST in Good Stories

- Bill Wake has offered six criteria (summarized by the acronym INVEST) that have proved useful when evaluating whether our stories are fit for their intended use or require some additional work.
- The INVEST criteria are Independent, Negotiable, Valuable, Estimatable, Small (sized appropriately), and Testable.

INVEST: Independent

- As much as is practical, user stories should be independent or at least only loosely coupled with one another.
- Stories that exhibit a high degree of interdependence complicate estimating, prioritizing, and planning.
- When applying the independent criteria, the goal is not to eliminate all dependencies, but instead to write stories in a way that minimizes dependencies.

INVEST: Independent

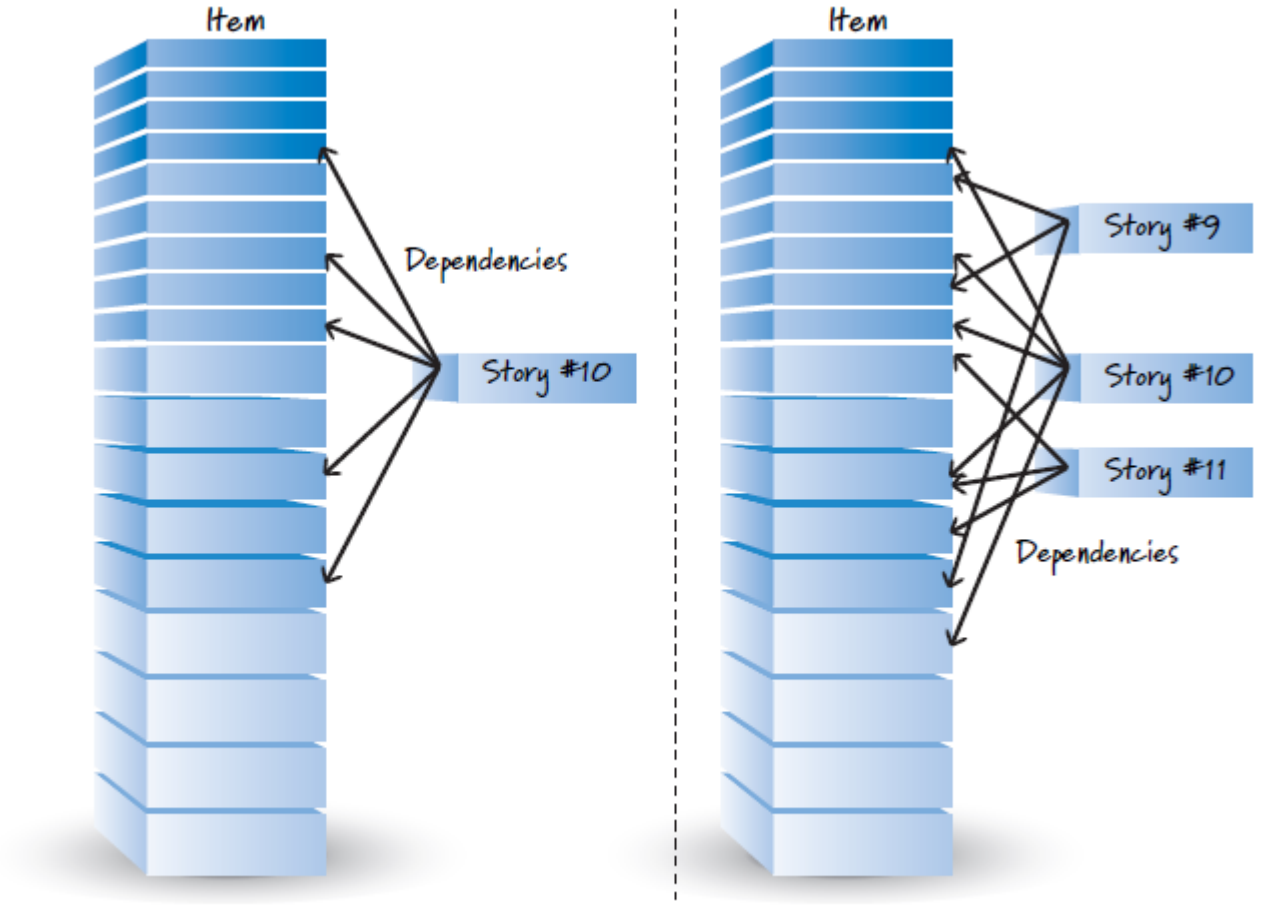


FIGURE 5.8 Highly dependent stories

INVEST: Negotiable

- The details of stories should also be negotiable. Stories are not a written contract in the form of an up-front requirements document.
- Instead, stories are placeholders for the conversations where the details will be negotiated.
- Good stories clearly capture the essence of what business functionality is desired and why it is desired.
- However, they leave room for the product owner, the stakeholders, and the team to negotiate the details.

INVEST: Negotiable

- A common example of where negotiability is violated is when the product owner tells the team how to implement a story.
- Stories should be about what and why, not how.
- When the how becomes non-negotiable, opportunities for the team to be innovative are diminished.
- The resulting innovation waste could have devastating economic consequences.

INVEST: Valuable

- Stories need to be valuable to a customer, user, or both.
- The fundamental problem with technical stories is that the product owner might not perceive any value in them.
- In practice, though, most technical stories should not be included in the product backlog.

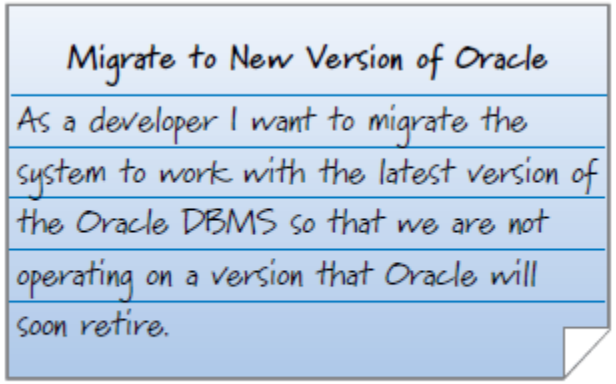


FIGURE 5.9 Example technical story

INVEST: Estimatable

- Stories should be estimatable by the team that will design, build, and test them.
- A large story that we plan to work on soon will need to be broken into a set of smaller stories.

INVEST: Small (sized appropriately)

- Stories should be sized appropriately for when we plan to work on them.
- Stories worked on in sprints should be small. If we're doing a several-week sprint, we want to work on several stories that are each a few days in size.
- If we have a two-week sprint, we don't want a two-week-size story, because the risk of not finishing the story is just too great.

INVEST: Testable

- Stories should be testable in a binary way—they either pass or fail their associated tests.
- Being testable means having good acceptance criteria (related to the conditions of satisfaction) associated with the story, which is the “confirmation” aspect of a user story.
- Without testable criteria, how would we know if the story is done at the end of the sprint?

Knowledge-Acquisition Stories

- Sometimes, we need to create a product backlog item that focuses on knowledge acquisition.
- Perhaps we don't have enough exploitable knowledge about the product or the process of building the product to move forward.
- Such exploration is known by many names: prototype, proof of concept, experiment, study, spike, and so on.

Knowledge-Acquisition Stories

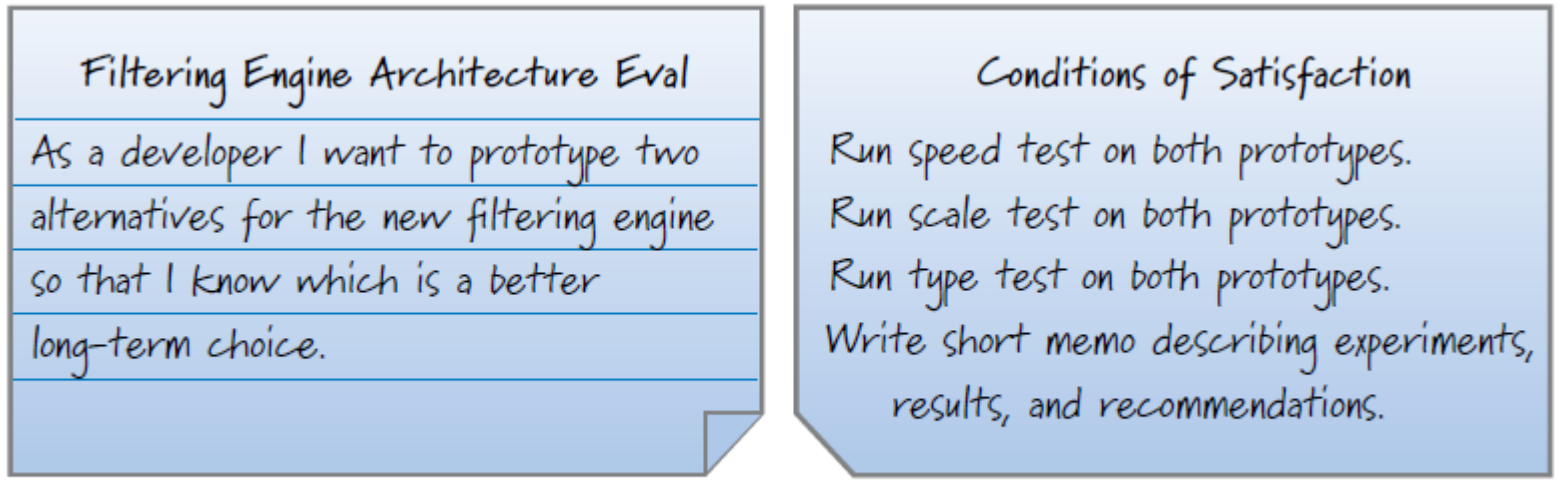


FIGURE 5.12 Knowledge-acquisition story

Story Mapping

- Story mapping is a technique popularized by Jeff Patton that takes a user-centric perspective for generating a set of user stories.
- The basic idea is to decompose high-level user activity into a workflow that can be further decomposed into a set of detailed tasks.
- Patton uses terms like activity, task, and subtask to describe the hierarchy inside a story map. To be consistent with the terminology we introduced earlier, we use epic, theme, and sprintable story.

Story Mapping

- At the highest level are the epics, representing the large activities of measurable economic value to the user—for example, the "Buy a Product" epic.
- Next, we think about the sequence or common workflow of user tasks that make up the epic (represented by themes—collections of related stories).
- We lay out the themes along a timeline, where themes in the workflow that would naturally occur sooner are positioned to the left of the ones that would occur later.
- For example, the "Search for Product" theme would be to the left of the "Manage Shopping Cart" theme.

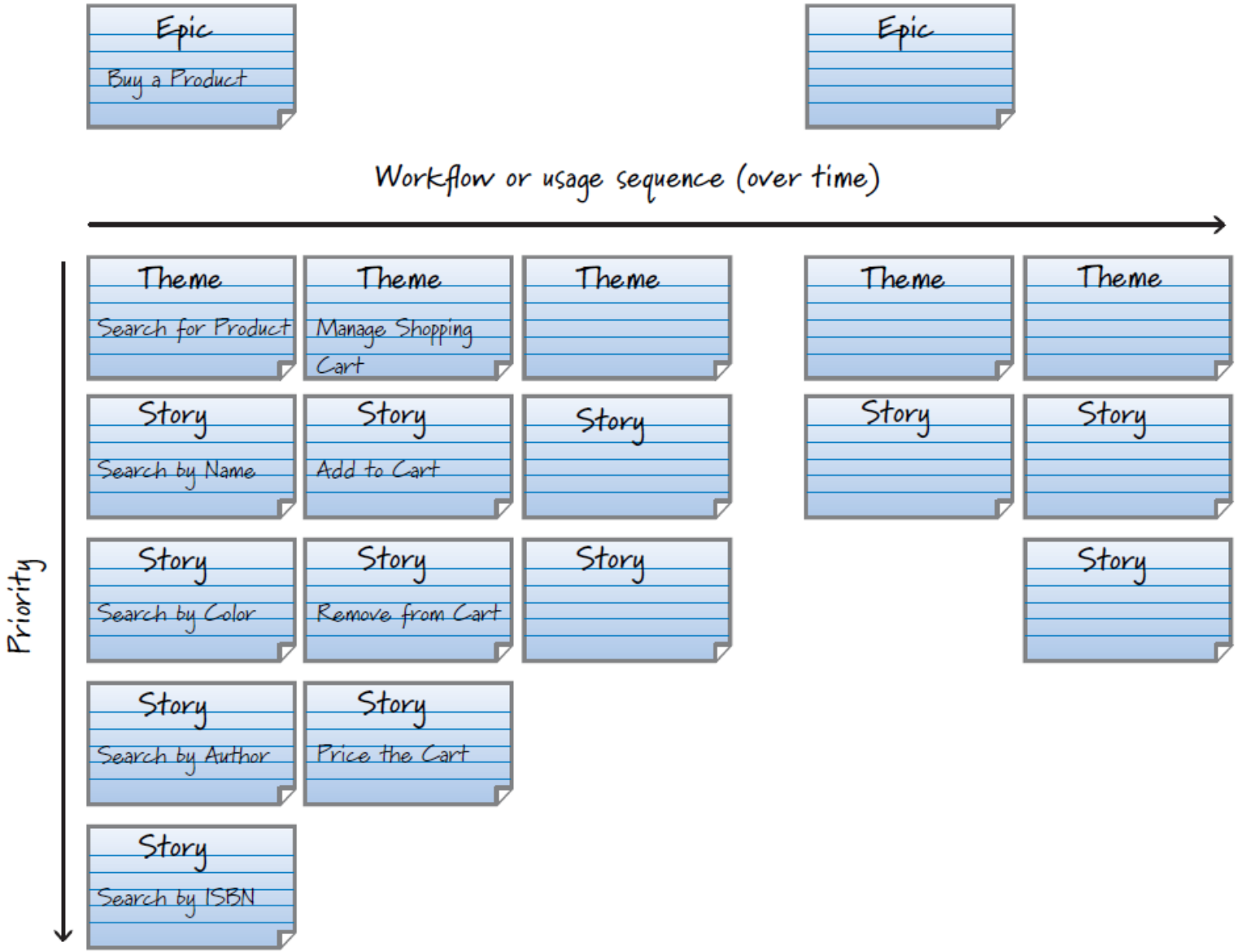
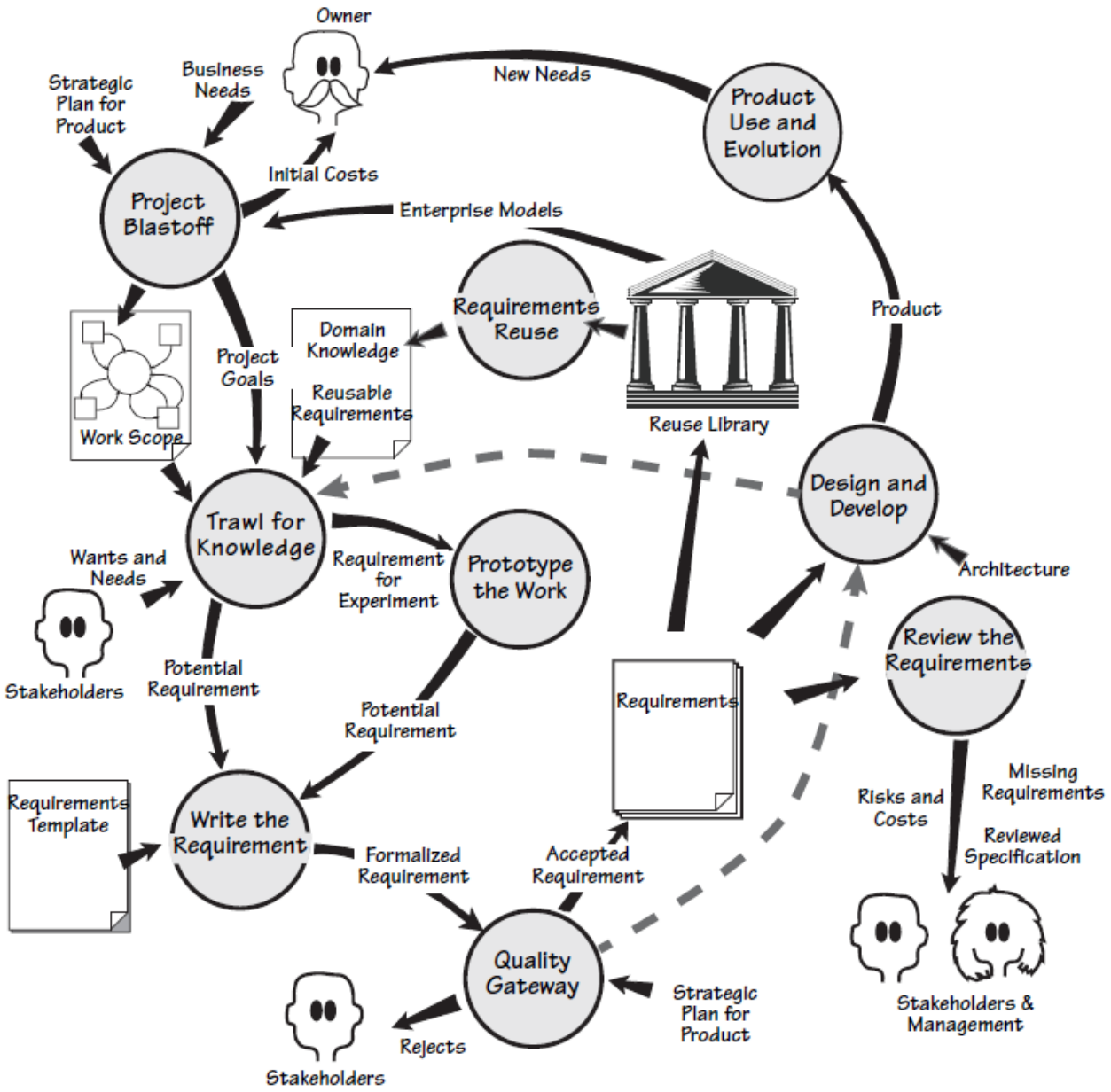


FIGURE 5.13 Story map

To Sum Up…

Figure 2.1

This map of the Volere Requirements Process shows the activities and their deliverables. We have used a stylized data flow notation. Each activity (the bubbles) and its deliverables (named arrows or documents) are explained in the text. The dotted lines represent how this process is used with iterative projects.



Next On

Take Away Points

- Advantage and limitations of each of the elicitation techniques:
  - Interviews
  - Questionnaires
  - Requirements Workshops
  - Ethnography
  - Prototyping

How to prepare user stories in SCRUM.

Further Reading

- Robertson S. and Robertson J, Mastering the Requirements Process: Getting Requirements Right
- K. Wiegers and J. Beatty, “Software Requirements (Developer Best Practices) 3rd edition”
- Kenneth Rubin, “Essential Scrum: A Practical Guide to the Most Popular Agile Process”
  - Chapter 5: Scrum and User Stories
- Mike Cohn, “User Stories Applied for Agile Software Development”

Further Reading

- Dean Leffingwell, “Agile Software Requirements”
  - Chapter 6: User Stories
  - Chapter 7: Stakeholders, User Personas, and user Experiences
- Pressman and Maxim, “Software Engineering”
  - Chapter 8: Understanding Requirements
- Ian Somerville, “Software Engineering”
  - Chapter 4: Requirements Engineering

Next Lecture

- Verification of functional requirements specification